MIL-C-37171
15 May 1975

MILITARY SPECIFICATION

# COOLANT AND CUTTING FLUID, LENS PREPARATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of petroleum oil based compounds used in optical lens processing machinery (see 6.1).

1.2 Classification. Coolant-cutting fluids shall be of the following types, as specified:

Type I - Water soluble concentrate.
Type II - Mineral oil compound.

## 2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect, on date of invitation for bids or request for proposals, form a part of this specification, to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-P-704 | Pail, Shipping, Steel (1 through 12 Gallon) |

STANDARDS

Federal

| | |
|---|---|
| Fed. Test Method: Std. No. 791 | Lubricants, Liquid Fuels, and Related Products; Methods of Testing. |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

FSC 6540

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS
(ASTM Test Methods and Specifications)

| | |
|---|---|
| D88 | Saybolt Viscosity |
| D92 | Flash and Fire Points by Cleveland Open Cup |
| D97 | Pour Point |
| D129 | Sulfur in Petroleum Products |
| D130 | Detection of Copper Corrosion by Copper Strip Tarnish Test |
| D287 | API Gravity of Crude Petroleum and Petroleum Products (Hydrometer Method) |
| D460 | Sampling and Chemical Analysis of Soaps and Soap Products |
| D808 | Chlorine in New and Used Petroleum Products |
| D1500 | Color of Petroleum Products (ASTM Color Scale) |
| D2270 | Viscosity Index |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material. Materials used shall be as follows:

3.1.1 Type I. Petroleum oil, emulsifers, coupling agents and additives such as detergents, rust inhibitors and bactericides.

3.1.2 Type II. Mineral oil base.

3.2 Physical and Chemical requirements. Each type shall be a stable homogenous preparation and shall conform to the applicable requirements specified in table I.

Table I. Physical and Chemical Requirements

| Properties | Requirements | |
|---|---|---|
| | Type I | Type II |
| Gravity, API, at 60°F, degrees | 12 to 14 | 39 to 41 |
| Viscosity, SUS at 100°F | 320 to 340 | 40 to 42 |
| Viscosity, SUS at 210°F | 55 to 65 | - |



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Table I. Physical and Chemical Requirements (Cont'd)

| Properties | Requirements | |
|---|---|---|
| | Type I | Type II |
| Viscosity index, minimum | 130 | -- |
| Flash Point, COC, $^{o}$F, min. | -- | 270 |
| Fire Point, COC, $^{o}$F, min. | -- | 310 |
| Pour Point, $^{o}$F, maximum | -- | 25 |
| Sulfur, percent (maximum) | 0.20 to 0.30 | 0.10 |
| Chlorine, percent, maximum | -- | 0.10 |
| Free fatty acid (as oleic) percent | 0.50 to 0.60 | ---- |
| Color, ASTM No., maximum | -- | 0.5 |
| Copper corrosion | -- | None |
| Odor | Non-offensive | |

3.2.1 Emulsion. (Type I only) The coolant concentrate shall require a minimum dilution of 20 parts water to one part of coolant, before use. The pH of the resulting emulsion shall be a maximum of 10.6.

3.2.2 Directions for use. The directions for use shall be plainly and indelibly printed on a label or printed directly on the can. The directions shall include: for Type I - instructions for dilution and shall state "For use in ophthalmic lens edging machines." For Type II - shall state "For use with Generator, Ophthalmic Lens, NSN 6540-00-782-6774, Ready for use, nothing to be added."

3.3 Toxicity. The coolants shall have no adverse effect on health of personnel in either concentrated or dilute form, when used for the intended purpose.

3.4 Workmanship. The coolant-cutting fluids shall be processed in accordance with high grade commercial practice and shall be suitable for intended purpose.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facitities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the date of testing, the test method, and individual test results.

4.2 Lot. A lot shall consist of an indefinite number of unit containers of identical size and type, offered for acceptance at one time and filled with a homogeneous mixture of one type of fluid manufactured in a single plant run with no change in ingredients.

4.3 Sampling.

4.3.1 Sampling for the examination of the preparation for delivery. A random sample of containers from each lot shall be taken in accordance with MIL-STD-105, as described in 4.6.

4.3.2 Sampling for tests. Perform sampling for tests in accordance with MIL-STD-105, inspection level S-3. Acceptance level shall be zero. The sample unit shall be the unit container.

4.4 Examination. The material shall be examined for appearance as described under workmanship and odor. Any nonconformance shall be cause for rejection.

4.5 Test procedures. Tests to determine conformance to the physical and chemical property requirements shall be conducted in accordance with methods of Federal Test Method Standard No. 791 or the American Society for Testing and Materials (ASTM) methods indicated in table II.



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Table II. Test methods

| Property | Fed. Std. 791, Method No. | ASTM Method |
|---|---|---|
| Color | 102.7 | D1500 |
| Gravity | 401.6 | D287 |
| Viscosity | 304.8 | D88 |
| Viscosity index | 9111.2 | D2270 |
| Flash and Fire Point | 1103.6 | D92 |
| Pour Point | 201.8 | D97 |
| Free fatty acid | ---- | D460 |
| Sulfur | 5202.12 | D129 |
| Chlorine | 5651.4 | D808 |
| Copper corrosion | 5313 | D130 |

4.5.1 pH value. Using any recognized standard make of pH apparatus, determine the pH value of the emulsion ( see 3.2.1)

4.6 Examination for preparation for delivery. An examination shall be made to determine whether the packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-3 and the acceptable quality level shall be 4.0 defects per hundred units.

Tabel III. Examination for Preparation for Delivery

| Examination | Defect |
|---|---|
| Containers | Not as specified. |
| Quantity | Quantity in cans not as specified. |
| Packing and Packaging Materials | Any component of can missing or damaged. |
| Workmanship | Inadequate application of components, such as incomplete closure of container, or containers distorted. |
| Markings | Omitted, incorrect, illegible, improper size, location, sequence, or method of application. |

5 PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One can containing five gallons of coolant of the type specified, constitutes one unit of issue.



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5.1.2 Level A. Five gallons of coolant (one unit of issue) of the type specified shall be contained in a steel pail conforming to the requirements of PPP-P-704, type 1, class 3. The can shall be provided with a self venting pour spout and having a screw cap closure. The interior of the pail shall be lined with a suitable coating to resist corrosion. The container, after lining, and the closure shall not interact physically or chemically with the contents. The leak test as specified in the appendix of PPP-P-704 shall apply.

5.1.3 Level C. The subject commodity shall be contained in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking. Marking shall be as specified in MIL-STD-129. In addition, the lot (control) number shall be shown.

6. NOTES

6.1 Intended use. Coolants and cutting fluids covered by this specification are intended for use in ophthalmic lens fabrication machines as described below:

Type I - The water soluble - emulsifiable concentrate is for use in lens cutting and edging machines.
Type II - The mineral oil compound is intended for use with ophthalmic lens generators (such as NSN 6540-00-782-6774).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type (see 1.2) and quantity desired.
c. Applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Catalog information. The coolant-cutting fluids covered by this specification are listed in the Federal Supply Catalog under the following stock numbers:-

| | National Stock Number | Item Identification |
|---|---|---|
| Type I | 6540-00-935-1137 | COOLANT AND CUTTING FLUID, LENS PREPARATION, WATER SOLUBLE, 5 gal. |
| Type II | 6540-00-999-0504 | COOLANT AND CUTTING FLUID, LENS PREPARATION, 5 gal. |

6.4 This specification does not include all types, classes and grades of the commodity indicated by the title of this specification or which are commercially available, but is intended to cover the types normally procured to meet Military requirements.



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Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
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